

11/8/2023

TASK - 2

generating design of commands other (conditional)

AIM:- TO perform (DDL) data definition language and (DML) data manipulation language commands

DDL commands:

- create → rename
- drop
- alter
- truncate

DML commands:

- Insert
- Update
- Delete
- SELECT

DDL Commands:

* Creating a Table by using create commands.

```
CREATE TABLE studentC
```

```
stu - id int,
```

```
stu - name varchar(30),
```

```
stu - department varchar(10),
```

```
stu - gender varchar(5),
```

```
stu - Ph. no int;);
```


After Alter table
desc student;

STU-ID	INT
STU-NAME	VARCHAR(30)
STU-DEPARTMENT	VARCHAR(30)
STU-GENDER	VARCHAR(5)
STU-PH-NO	INT
STU-DEPART-ID	INT

After Inserting into table

SELECT * FROM student;

S-no	STU-ID	Stu-Name	STU-department	STU-gender
1	28487	Shravan	CSE	male
2	29511	Krishna	CSE	male

STU-Ph-No	STU-depart-id
8688078	1225
432578	1425

* Using ALTER command we can add or remove the column

SYNTAX:

ALTER table student ADD column stu-dept-id int;

DROP :

* DROP command is used to drop the table completely

DROP TABLE student

TRUNCATE

* TRUNCATE command is used to remove all data but keep structure

TRUNCATE TABLE student

DML COMMANDS

INSERT

* Insert command is used to insert the values to the table

INSERT INTO student VALUES

(28487, 'Jagannadham', 'Cse', 'male', 18688078)

(29511, 'Krishna', 'Cse', 'male', 432578)

UPDATE

* UPDATE command is used to update the existing records.

UPDATE student

SET stu-name = 'Mohith'

WHERE stu-id = 28800.

After Updating the table.

SELECT * FROM student

S.No	STU-id	STU-Name	STU-department	STU-gender
1	28487	Jagannadham	CSE	male
2	29511	Krishan	CSE	male
		STU-phone-No	STU-depart id	
		8688078	1225	
		432578	1425	

DELETE:

* Delete command is used delete a record

DELETE FROM Student

WHERE Stu-ID = 28800;

SELECT

* select & command is used to retrieve the records from table.

SELECT * FROM Student

Sr No	STU-ID	STU-Name	STU-department	STU-Gender	STU-Ph-No
1	28487	Jagannadham	CSE	male	8688078

EX:

CREATE TABLE Students C

~~Roll-NO~~ INT,

Name VARCHAR(30),

Age INT

Course VARCHAR(30));

ALTER TABLE Students ADD

Email VARCHAR(50);

INSERT INTO Students VALUES

(1) ('Jagannadham', 19, 'Btech', 'jagannadham@gmail.com

(2) ('Roy', 20, 'Btech', 'apple@gmail.com)

(3) ('Joy', 21, 'Btech', 'joy@gmail.com');

UPDATE Students

SET Email = utu28487@gmail.com

WHERE Roll-NO = 1;

DELETE FROM Students
WHERE Roll - no = 2;

SELECT * FROM Students ;

Sno	Roll No	Name	Age	Course	Email
1	1	Jagannadham	19	B Tech	vtu28487@gmail.com
2	3	Joy	21	B-Tech	joy@gmail-com

SELECT Name FROM Students;

Sno	Name
1	Jagannadham
2	Joy.

SELECT * FROM Student S
WHERE Name = 'Jagannadham';

sno	Roll - no	name	Age	course	Email
1	1	Jagannadham	19	BTech	vtu28487@gmail-com

VELTECH	
EX No.	2.1
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	4
RECORD (5)	-
TOTAL (20)	14
SIGN WITH DATE	

RESULT:

All the DDL and DML commands are successfully executed

4/8/2025 TASK (2.2)

DDL and DML commands with constraints

AIM: TO perform DDL and DML commands with constraints in SQL

CONSTRAINTS

- NOT NULL
- UNIQUE
- PRIMARY KEY
- FOREIGN KEY
- CHECK
- DEFAULT

NOT NULL

It ensures a column cannot store NULL values

SYNTAX:

```
CREATE TABLE TABLE_NAME (  
    COLUMNNAME DATA TYPE NOT NULL;
```

UNIQUE

It ensure all values in a column are unique

SYNTAX

```
CREATE TABLE table name (  
    column name datatype UNIQUE);
```

PRIMARY KEY

It is the combination of NOT NULL & UNIQUE

FOREIGN KEY :

It ensure values in one table match values in another table

SYNTAX

FOREIGN KEY (column name) REFERENCE
another table name (column name)

DEFAULT:

IT provides a default value for a column when none is specified.

Example

```
CREATE TABLE DEPARTMENT  
DEPT-ID INT PRIMARY KEY,  
DEPT-NAME VARCHAR(20) UNIQUE NOT NULL;
```

```
CREATE TABLE STUDENT  
STU-ID INT PRIMARY KEY,  
STU-NAME VARCHAR(30) NOT NULL,  
STU-DEPARTMENT INT DEFAULT 101,  
STU-GENDER VARCHAR(6)  
CHECK (STU-GENDER IN ('MALE',  
                        'FEMALE'));
```

```
STU-PH100 BIGINT UNIQUE,  
FOREIGN KEY (STU-DEPARTMENT) REFERENCES  
DEPARTMENT (DEPT-ID);
```


INSERT INTO DEPARTMENT VALUES

(101, 'CSE');
(102, 'ECE');
(103, 'IT');

INSERT INTO STUDENT VALUES

(1, 'RAVI', 102, 'MALE', 9876543210);
(2, 'ANITA', 'FEMALE', 9816501234);

SELECT * FROM DEPARTMENT;

SELECT * FROM STUDENT;

ALTER TABLE STUDENT

ADD ~~STU-EMAIL~~ VARCHAR(60) DEFAULT
~~cvu@tech@gmail.com;~~

S-NO	STU-ID	STU-NAME	STU- DEPARTMENT	STU- GENDER	STU-PH	STU-EMAIL
1	1	RAVI	102	MALE	9876543210	NULL
2	2	ANITA	101	FEMALE	9816501234	NULL

UPDATE STUDENT

SET ~~STU-DEPARTMENT~~ = 103

WHERE ~~STU-NAME~~ 'ANITA'

~~DELETE FROM DEPARTMENT~~

WHERE ~~DEPT-ID~~ = 103;

INSERT INTO STUDENT VALUES;

(2, 'ANITA', 'FEMALE', 9876501234);

(3, 'Shreasha', 'FEMALE', 9876501320);

SELECT * FROM STUDENT - Before performing

ALTER COMMAND

S.No	STU-ID	STU-NAME	STU-DEPARTMENT	STU-GENDER	STU-PHNO
1	1	RAVI	102	MALE	987654320
2	2	ANITA	101	FEMALE	9876501234

SELECT * FROM DEPARTMENT - - Before performing
ALTER COMMAND

S.No	STU-ID				
1	101	CSE			
2	102	ECE			
3	103	IT			

SELECT * FROM STUDENT - - after performing update
Command

	STU-ID	STU-NAME	STU-DEPARTMENT	STU-GENDER	STU-PHNO	STU-EMAIL
1	1	RAVI	102	MALE	9876543210	NULL
2	2	ANITA	103	FEMALE	9876501234	NULL

SELECT * FROM STUDENT - - after performing delete
Command

S.No	STU-ID	STU-NAME	STU-DEPARTMENT	STU-PHNO	STU-EMAIL
1	1	RAVI	102	9876543210	NULL

SELECT * FROM STUDENTS - AFTER inserting values.

SNO	STU-NAME	STU- ^{DEPART} MENT	STU-GENDER	STU-PHNO	STU-EMAIL
1	RAVI	102	MALE	9876547210	Voltoth@NLU
2	ANITA	101	FEMALE	9876501234	Voltoth@gmail.co
3	Shreshtha	101	FEMALE	9876501230	Voltoth@gmail.com

SELECT * FROM DEPARTMENT: After inserting values.

	DEPT-ID	DEPT-NAME
1	101	CSE
2	102	ECE
3	103	IT



DROP TABLE DEPARTMENT;

--- ERROR ---

could not drop object DEPARTMENT because it is referenced by a FOREIGN KEY constraint

~
To solve this first we have to drop student table after we have to drop department table.

VEL TECH	
EX No.	2.2
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	15
SIGN WITH DATE	

4/8/3

RESULT: All the DDL and DML commands are with constraints are performed and executed successfully.